

#This code will be used to demonstrate lemma 7.10. we demonstrate the existence of the curve γ in region R2.

#Construct the polynomial R from Theorem 7.1.2.

$$R := (2 \cdot w^3 \cdot z^3)^2 - (w^3 + z^3)^2 \cdot \left(z^2 + \frac{y \cdot x}{2} - \frac{y^2}{4} \right)^3$$

$$R := 4 w^6 z^6 - (w^3 + z^3)^2 \left(z^2 + \frac{1}{2} y x - \frac{1}{4} y^2 \right)^3 \quad (1)$$

#Apply the rational parametrization constructed in this chapter 4..

$$\text{parametrization} := \left\{ \begin{array}{l} w = a + \frac{1}{a}, \\ z = b + \frac{1}{b}, \\ x = a - \frac{1}{a} + b - \frac{1}{b}, \\ y = a - \frac{1}{a} - b + \frac{1}{b} \end{array} \right\} :$$

simplify(numer(expand(subs(parametrization, R)))) ;

denom(expand(subs(parametrization, R))) ;

$$\begin{aligned} & - (a - b)^2 (b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} \\ & - 102 b^{11} - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} \\ & + 1380 b^{15} + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} \\ & - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} \\ & - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} \\ & - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} \\ & + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} \\ & + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 \\ & + 198 b^2 + 15) a^{12} - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} \\ & + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} \\ & - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} \\ & - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 \\ & + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 \\ & + 10) a^8 + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 \\ & + 74 b^3) a^7 + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 \\ & + 145 b^4) a^6 + (24 b^{15} - 96 b^{13} - 762 b^{11} + 1884 b^9 + 1380 b^7 + 252 b^5) a^5 + (3 b^{14} \\ & + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^4 + (-18 b^{13} - 102 b^{11} + 162 b^9 + 74 b^7) a^3 \end{aligned}$$

$$+ (15 b^{12} + 6 b^{10} + 30 b^8) a^2 + (-6 b^{11} + 8 b^9) a + b^{10})$$

$$64 a^{12} b^{12}$$

(2)

#Define r4

$$\begin{aligned} r4 := & (b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} \\ & - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} + 1380 b^{15} \\ & + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\ & - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} \\ & - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} \\ & - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} \\ & - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\ & - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{12} \\ & - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 \\ & + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} \\ & - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\ & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 + 3 b^2 (b^{16} - 580 b^{14} \\ & - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^8 + (24 b^{17} - 132 b^{15} \\ & - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 + 74 b^3) a^7 + (-294 b^{16} - 1740 b^{14} \\ & - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^6 + (24 b^{15} - 96 b^{13} - 762 b^{11} \\ & + 1884 b^9 + 1380 b^7 + 252 b^5) a^5 + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^4 + (-18 b^{13} \\ & - 102 b^{11} + 162 b^9 + 74 b^7) a^3 + (15 b^{12} + 6 b^{10} + 30 b^8) a^2 + (-6 b^{11} + 8 b^9) a \\ & + b^{10}) : \end{aligned}$$

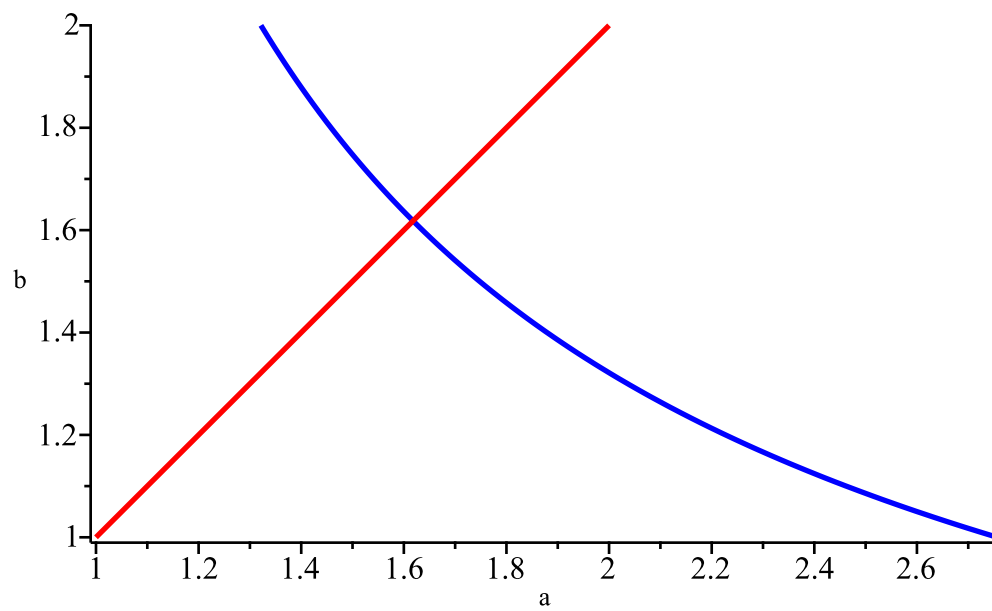
#Plot the curve r4=0 in region A1.

$$r := a - b :$$

with(plots) :

```
implicitplot([r4=0, r=0], a=1 .. 3, b=1 .. (2),
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, red],
  labels=["a", "b"],
  title="Graphic for r4=0 in A1");
```

Graphic for $r_4=0$ in A1



#Apply the Möbius transformation on the rectangle $A1 = [v, 3) \times [1, v)$ constructing r^4

$$v := \frac{1 + \sqrt{5}}{2} :$$

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{v + 3 \cdot c}{1 + c}, \\ b = \frac{1 + v \cdot d}{1 + d} \end{array} \right\} :$$

$\hat{r}^4 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, r^4))), c) :$

#Compute the degree of $\hat{r}^4$ in c

$\text{degree}(\hat{r}^4, c);$

22

(3)

#Define the coefficients $c_i = \text{coeff}(\hat{r}^4, c, i)$
 $\text{seq}(\text{evalf}(\text{assign}(\text{cat}('c', i), \text{coeff}(\hat{r}^4, c, i))), i = 0 .. 22);$

#Compute the all degree of c_i in d
 $\text{seq}(\text{degree}(\text{eval}(\text{cat}('c', i)), d), i = 0 .. 22);$

21, 22

(4)

#From here we will analyze the signal variation of all coefficients c_i

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{22}, d, i))), i = 22 .. 0, -1);$

1, 1

(5)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{21}, d, i))), i = 22 .. 0, -1);$

1, 1

(6)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{20}, d, i))), i = 22 .. 0, -1);$

1, 1

(7)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{19}, d, i))), i = 22 .. 0, -1);$

$\text{fsolve}(c_{19} = 0, d, 0 .. \text{infinity});$

1, -1

0.01248617890

(8)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{18}, d, i))), i = 22 .. 0, -1);$

$\text{fsolve}(c_{18} = 0, d, 0 .. \text{infinity});$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1

0.05747338158

(9)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{17}, d, i))), i = 22 .. 0, -1);$

$$\begin{aligned} &fsolve(c17=0, d, 0 \dots \text{infinity}); \\ &\quad 1, -1, -1, -1 \\ &\quad 0.1076291361 \end{aligned} \tag{10}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c16, d, i))), i = 22 .. 0, -1); \\ & fsolve(c16 = 0, d, 0 .. infinity); \\ & \quad 1, -1, -1, -1, -1 \\ & \quad \quad \quad 0.1639320599 \end{aligned} \tag{11}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c15, d, i))), i = 22 \dots 0, -1); \\ & fsolve(c15 = 0, d, 0 \dots \text{infinity}); \end{aligned}$$

$$\begin{aligned} & 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1 \\ & 0.2276219640 \end{aligned} \tag{12}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c14, d, i))), i = 22 .. 0, -1); \\ & fsolve(c14 = 0, d, 0 .. infinity); \\ & \quad 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1 \\ & \quad 0.3002932460 \end{aligned} \tag{13}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c13, d, i))), i = 22 .. 0, -1); \\ & fsolve(c13 = 0, d, 0 .. infinity); \\ & \quad 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1 \\ & \quad \quad \quad 0.3840313506 \end{aligned} \tag{14}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c12, d, i))), i = 22 .. 0, -1); \\ & fsolve(c12 = 0, d, 0 .. infinity); \\ & \quad 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \\ & \quad 0.4816174281 \end{aligned} \tag{15}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c11, d, i))), i = 22 .. 0, -1); \\ & fsolve(c11 = 0, d, 0 .. infinity); \\ & \quad 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \\ & \quad 0.5968445961 \end{aligned} \tag{16}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c10, d, i))), i = 22 .. 0, -1); \\ & fsolve(c10 = 0, d, 0 .. infinity); \\ & \quad 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \\ & \quad \quad \quad 0.7350239286 \end{aligned} \tag{17}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c9, d, i))), i = 22 .. 0, -1); \\ & fsolve(c9 = 0, d, 0 .. infinity); \\ & \quad 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \\ & \quad \quad \quad 0.9038277452 \end{aligned} \tag{18}$$

```
seq(sign(evalf(coeff(c8,d,i))), i=22..0,-1);  
fsolve(c8=0,d,0..infinity);  
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1
```

$$1.114765340 \quad (19)$$

```
seq(sign( evalf( coeff( c7, d, i ) ) ), i = 22 .. 0, -1);
fsolve( c7 = 0, d, 0 .. infinity);
1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1
```

$$1.385923600 \quad (20)$$

```
seq(sign( evalf( coeff( c6, d, i ) ) ), i = 22 .. 0, -1);
fsolve(c6 = 0, d, 0 .. infinity);
1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1
```

$$1.747448203 \quad (21)$$

```
seq(sign( evalf( coeff( c5, d, i ) ) ), i = 22 .. 0, -1);
fsolve( c5 = 0, d, 0 .. infinity);
1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1
```

$$2.253602253 \quad (22)$$

```
seq(sign(evalf(coeff(c4,d,i))), i=22..0,-1);  
fsolve(c4=0,d,0..infinity);
```

1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,
-1

$$3.012912810 \quad (23)$$

```
seq(sign( evalf(coeff(c3,d,i) ) ), i=22 .. 0,-1 );
fsolve(c3=0, d, 0 .. infinity);
```

$$4.278610483 \quad (24)$$

```
seq(sign( evalf( coeff( c2, d, i ) ), i = 22 .. 0, -1 );
fsolve(c2=0, d, 0 .. infinity);
```

1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,
-1, -1

$$6.810387750 \quad (25)$$

[illegible]

$$14.40670717 \quad (26)$$

$$\begin{aligned} & seq(sign(evalf(coeff(c0, d, i))), i = 21 .. 0, -1); \\ & -1, \\ & \quad -1, -1 \end{aligned} \quad (27)$$

restart;

#Define r4

$$\begin{aligned}
 r4 := & (b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} \\
 & - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} + 1380 b^{15} \\
 & + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\
 & - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} \\
 & - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} \\
 & - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} \\
 & - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\
 & - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{12} \\
 & - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 \\
 & + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} \\
 & - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\
 & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 + 3 b^2 (b^{16} - 580 b^{14} \\
 & - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^8 + (24 b^{17} - 132 b^{15} \\
 & - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 + 74 b^3) a^7 + (-294 b^{16} - 1740 b^{14} \\
 & - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^6 + (24 b^{15} - 96 b^{13} - 762 b^{11} \\
 & + 1884 b^9 + 1380 b^7 + 252 b^5) a^5 + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^4 + (-18 b^{13} - 102 b^{11} + 162 b^9 + 74 b^7) a^3 \\
 & + (15 b^{12} + 6 b^{10} + 30 b^8) a^2 + (-6 b^{11} + 8 b^9) a + b^{10}) :
 \end{aligned}$$

#Apply the Möbius transformation on the rectangle $A2 = [1, v) \times [1, v)$ constructing $\hat{r}4$

$$v := \frac{1 + \sqrt{5}}{2} :$$

$$\text{moebius} := \left\{ \begin{aligned} a &= \frac{1 + v \cdot c}{1 + c}, \\ b &= \frac{1 + v \cdot d}{1 + d} \end{aligned} \right\} :$$

$$\hat{r}4 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, r4))), c) :$$

#Compute the degree of $\hat{r}4$ in c

$$\text{degree}(\hat{r}4, c);$$

22

(28)

#Define the coefficients $ci = \text{coeff}(\hat{r}4, c, i)$

for i from 0 to 22 do

```
#Compute the all degree of ci in d
seq(degree(c[i], d), i = 0..22);
```

(29)

[illegible]

restart;

#Define r4

$$\begin{aligned}
 r4 := & (b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} \\
 & - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} + 1380 b^{15} \\
 & + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\
 & - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} \\
 & - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} \\
 & - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} \\
 & - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\
 & - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{12} \\
 & - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 \\
 & + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} \\
 & - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\
 & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 + 3 b^2 (b^{16} - 580 b^{14} \\
 & - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^8 + (24 b^{17} - 132 b^{15} \\
 & - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 + 74 b^3) a^7 + (-294 b^{16} - 1740 b^{14} \\
 & - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^6 + (24 b^{15} - 96 b^{13} - 762 b^{11} \\
 & + 1884 b^9 + 1380 b^7 + 252 b^5) a^5 + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^4 + (-18 b^{13} - 102 b^{11} + 162 b^9 + 74 b^7) a^3 + (15 b^{12} + 6 b^{10} + 30 b^8) a^2 + (-6 b^{11} + 8 b^9) a \\
 & + b^{10}) :
 \end{aligned}$$

#Apply the Möbius transformation on the rectangle $A3 = [3, +\infty) \times [1, v)$ constructing $\hat{r}4$

$$v := \frac{1 + \sqrt{5}}{2} :$$

$$\text{moebius} := \left\{ \begin{array}{l} a = 3 + c, \\ b = \frac{1 + v \cdot d}{1 + d} \end{array} \right\} :$$

$$\hat{r}4 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, r4))), c) :$$

#Compute the degree of $\hat{r}4$ in c

$$\text{degree}(\hat{r}4, c);$$

22

(32)

#Define the coefficients $ci = \text{coeff}(\hat{r}4, c, i)$

for i from 0 to 22 do

```

    c[i] := coeff( r4, c, i);
od:

```

#Compute the all degree of c_i in d

```
seq( degree( c[i], d ), i = 0 .. 22 );
```

[illegible]

(33)

#From here we will analyze the signal variation of all coefficients c_i

$$seq(seq(sign(evalf(coeff(c[j], d, i))), i = 22..0, -1), j = 22..0, -1);$$
[illegible][illegible]

restart;

#Define r4

$$\begin{aligned} r4 := & (b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} \\ & - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} + 1380 b^{15} \\ & + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\ & - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} \\ & - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} \\ & - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} \\ & - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\ & - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{12} \\ & - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 \\ & + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} \\ & - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\ & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 + 3 b^2 (b^{16} - 580 b^{14} \\ & - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^8 + (24 b^{17} - 132 b^{15} \\ & - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 + 74 b^3) a^7 + (-294 b^{16} - 1740 b^{14} \\ & - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^6 + (24 b^{15} - 96 b^{13} - 762 b^{11} \\ & + 1884 b^9 + 1380 b^7 + 252 b^5) a^5 + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^4 + (-18 b^{13} \\ & - 102 b^{11} + 162 b^9 + 74 b^7) a^3 + (15 b^{12} + 6 b^{10} + 30 b^8) a^2 + (-6 b^{11} + 8 b^9) a \\ & + b^{10}) : \end{aligned}$$

#Apply the transformation on the rectangle $A4 = [v, +\infty) \times [v, +\infty)$ constructing $r4$

$$v := \frac{1 + \sqrt{5}}{2} :$$

$$\text{moebius} := \{ a = v + c, \\ b = v + d \} :$$

$$r4 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, r4))), c) :$$

#Compute the degree of $r4$ in c

$$\text{degree}(r4, c);$$

22

(35)

#Define the coefficients $ci = \text{coeff}(r4, c, i)$

for i **from** 0 **to** 22 **do**

$$c[i] := \text{coeff}(r4, c, i);$$

od:

```
seq( degree( c[i], d ), i = 0 .. 22 );
```

(36)

#From here we will analyze the signal variation of all coefficients c_i

$$seq(seq(sign(evalf(coeff(c[j], d, i))), i = 22..0, -1), j = 22..0, -1);$$
[illegible][illegible]